

★ WEEK 1 · PART 1 · PRACTICE

Returns, Worked by Hand

Simple and Log Returns, and Why Gains Don't Cancel Losses

Two views of one move, and the volatility drag

Part 1's foundations, with every digit shown. We compute simple and log returns from a price path - watching log returns add while simple returns multiply - then confront the volatility drag, where the arithmetic mean flatters and the geometric mean tells the truth.

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PART 1 · APPLICATIONS

The Unit Everything Is Measured In

Case A: compute simple and log returns from a two-day price path, and see why +10% then -10% leaves you down.
Case B: the arithmetic mean says break-even while the geometric mean shows a loss - the volatility drag, equal to about half the variance. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

We model returns, not prices

Returns are comparable and better behaved than raw prices. Simple returns compound multiplicatively; log returns add through time.

2

Simple and log are two views of one move

An exact, invertible relationship: the log return is the log of one plus the simple return. Close for small moves, not identical.

3

The geometric mean tells the truth

The arithmetic mean overstates compound growth. The gap - the volatility drag - is about half the variance.

Everything here is a calculator exercise: one returns table, one volatility drag.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a price path or a pair of returns.
- Compute both kinds, or both means.
- Then read what really happened.

What you should watch

- How log returns add and simple returns multiply.
- Why equal gains and losses don't cancel.
- How the drag equals half the variance.

Simple and Log Returns

Two views of one price path

A stock goes from 100 to 110, then to 99. We compute both the simple and the log return each day, and watch two facts emerge: log returns add up to the total log return, while gross simple returns multiply - and a +10% followed by a -10% leaves you at 99, not 100. Every digit is shown.

The price path

- Starts at 100.
- Up to 110, a 10% gain.
- Down to 99, a 10% loss.

What we'll do by hand

- Compute simple and log returns each day.
- Add the logs; multiply the gross simples.
- See where you actually end up.

The Two Definitions

Simple, gross, and log

$$R_t = \frac{P_t}{P_{t-1}} - 1 \quad (\text{simple}), \quad r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t) \quad (\text{log})$$

The simple return

- The percentage change over the period.
- Intuitive, and what most people quote.
- It compounds by multiplying.

The log return

- The log of the gross return.
- The modeler's default unit.
- It compounds by adding.

Both Returns, Day by Day

Follow every digit

day	price	simple R	log r
start	100	-	-
day 1	110	+10.00%	+9.53%
day 2	99	-10.00%	-10.54%
total	99	-1.00%	-1.005%

Day one is a clean +10%; day two is -10% off the higher base of 110, which costs 11 points. The simple returns don't sum to the total, but the log returns do: 9.53% minus 10.54% is -1.005%, exactly the log of 99 over 100.

Logs Add, Simples Multiply

Why +10% and -10% don't cancel

$$\ln(1.10) + \ln(0.90) = 0.0953 - 0.1054 = -0.01005 = \ln \frac{99}{100} \quad (\text{logs add; } +10\%, -10\% \rightarrow 99)$$

The log side

- The two log returns simply add.
- 9.53% minus 10.54% is -1.005%.
- That equals the total log return.

The simple side

- The gross returns multiply: 1.1 times 0.9.
- That's 0.99 - a total of -1%.
- You end at 99, not back at 100.

Two Views, One Move

Left, the price path: up 10% to 110, then down 10% off that higher base, landing at 99 - below where it started. Right, the daily returns: simple and log sit close together for these small moves but are never exactly equal, the log always a touch smaller on the way up and larger in magnitude on the way down.

Case A: two views of the same two moves



Why Log Is the Default

The modeler's convenience

Why modelers prefer logs

- They add across time, so multi-period is easy.
- They're symmetric around zero.
- They pair naturally with the normal distribution.

When simple is right

- For actual portfolio percentage changes.
- For quoting a client's realized gain.
- The two agree closely for small moves.

The Volatility Drag

When the average lies

Consider a return of +20% followed by -20%. The arithmetic mean is exactly zero, suggesting break-even - but you actually end at 96, down 4%. The geometric mean captures this: it's negative. We compute both, measure the gap - the volatility drag - and confirm it equals about half the variance.

The two returns

- Up 20%, then down 20%.
- Arithmetic mean: exactly zero.
- But the wealth path disagrees.

What we'll do by hand

- Compute the geometric mean.
- Measure the drag against the arithmetic.
- Match it to half the variance.

Arithmetic vs Geometric

Two ways to average returns

$$\bar{R}_{\text{arith}} = \frac{1}{n} \sum R_t, \quad \bar{R}_{\text{geo}} = \left(\prod (1 + R_t) \right)^{1/n} - 1 \leq \bar{R}_{\text{arith}}$$

The arithmetic mean

- Just the simple average of the returns.
- Answers: what to expect next period.
- But it overstates realized growth.

The geometric mean

- The constant rate that reproduces end wealth.
- Always less than or equal to the arithmetic.
- It's the honest compound rate.

The +20% / -20% Trap

Zero average, real loss

$$+20\%, -20\% : \bar{R}_{\text{arith}} = 0, \text{ but } 1.2 \times 0.8 = 0.96 \Rightarrow \bar{R}_{\text{geo}} = -\mathbf{2.02\%}$$

The arithmetic view

- Plus 20 and minus 20 average to zero.
- It looks like you broke even.
- You did not.

The geometric view

- 1.2 times 0.8 is 0.96 - down 4%.
- Per period, that's minus 2.02%.
- The wealth path ends at 96.

The Drag Is Half the Variance

And real vs nominal vs excess

$$\text{drag} = \bar{R}_{\text{arith}} - \bar{R}_{\text{geo}} = 0.0202 \approx \frac{1}{2}\sigma^2 = 0.02 \quad (\bar{R}_{\text{geo}} \approx \bar{R}_{\text{arith}} - \frac{1}{2}\sigma^2)$$

The drag

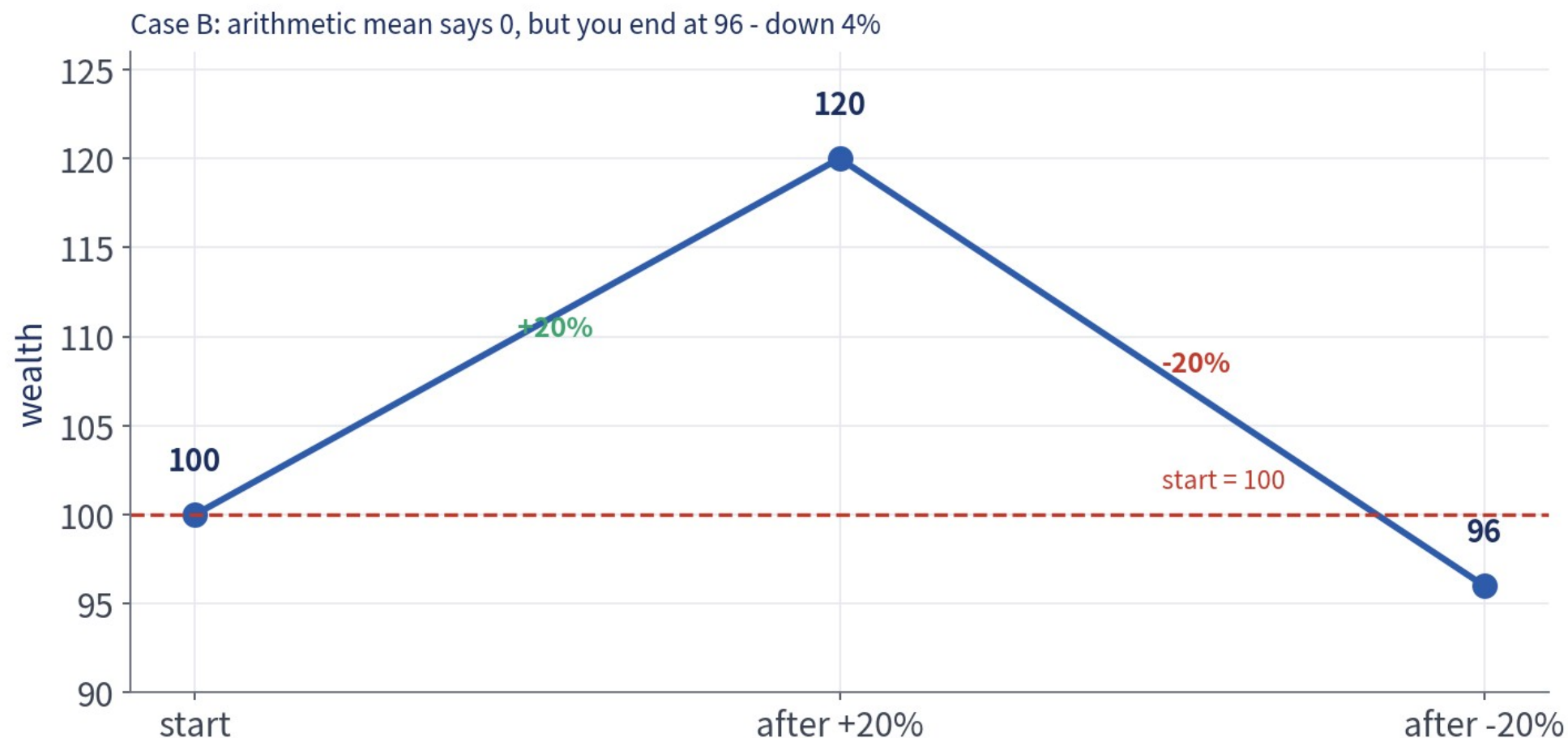
- Arithmetic minus geometric is 0.0202.
- Half the variance is 0.02.
- They match - drag is about half the variance.

Three adjustments

- Real: strip out inflation - 8% over 3% is 4.85%.
- Excess: subtract the risk-free - 8% minus 2% is 6%.
- Compare returns only after these.

Zero on Average, Down in Reality

Start at 100, gain 20% to 120, then lose 20% off that higher base - and land at 96, not 100. The arithmetic mean of the two returns is zero, but the wealth path tells the truth: a 4% loss. This gap between the average return and the realized growth is the volatility drag.



Comparing Returns Fairly

The adjustments that matter

Why the drag matters

- Volatile strategies lose to the drag.
- Quote the geometric mean for realized growth.
- The wider the swings, the bigger the gap.

Real, nominal, excess

- Nominal returns ignore inflation.
- Real returns strip it out - the true purchasing power.
- Excess returns subtract the risk-free rate.

The Returns Toolkit

The unit everything is built on

Returns are the foundation of everything that follows. Convert prices to returns, choosing log returns when you'll model and sum across time, simple returns when you'll compound a portfolio. Average with the geometric mean for honest realized growth, remembering the volatility drag of half the variance. And adjust for inflation and the risk-free rate before comparing. Get the unit right, and every later model inherits a clean input.

The returns (Case A)

- Simple +10%, -10%; log +9.53%, -10.54%.
- Logs add to -1.005%; simples multiply to -1%.
- End at 99, not 100.

The drag (Case B)

- Arithmetic mean 0, geometric -2.02%.
- Drag 0.0202, about half the variance.
- The average lied; the wealth path didn't.

Reproduce the Returns and the Drag

Direct Claude Code to check your arithmetic

GOAL Compute both returns from a price path, then the arithmetic and geometric means.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - returns:

- * for prices `[100, 110, 99]`, print the simple and log return each day;
- * confirm the logs sum to $\ln(99/100)$ and the gross simples multiply to 0.99.

CASE B - the drag:

- * for returns `[+0.20, -0.20]`, print the arithmetic and geometric mean;
- * confirm the drag is about half the variance.

Then plot the price path and the +20/-20 wealth path.

Check against the slides

- logs: +9.53%, -10.54% -> -1.005%.
- gross: $1.1 \times 0.9 = 0.99$.
- arith 0, geo -2.02%.
- drag 0.0202 ~ 0.02.

Deliverable: `part1_practice.py` whose printed numbers match the returns table and the drag.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do log returns compound over time?

**They add; gross
simple returns
multiply**

Q2 Why don't +10% and -10% cancel?

**The loss is off a
higher base - you end
at 99**

Q3 How big is the volatility drag?

**About half the
variance of the
returns**

All three come from the returns table, the multiplicative compounding, and the volatility drag.

Three Pitfalls to Avoid

Where returns trip people

Common mistakes

- "A +10% then -10% gets you back to even."
- "The arithmetic mean is the realized growth."
- "Log and simple returns are the same."

The reality

- The loss is off a higher base - you end down.
- Use the geometric mean; mind the drag.
- They agree only for small moves - not exactly.

Two Calculations to Remember

Part 1 applications in one screen

1

Log returns add; gains don't cancel losses

On the path 100 to 110 to 99, the log returns of +9.53% and -10.54% summed to -1.005%, exactly the log of 99 over 100, while the gross simples multiplied to 0.99. A +10% then -10% ends at 99 - the loss falls on a higher base.

2

The geometric mean tells the truth

For +20% then -20%, the arithmetic mean was zero but the wealth ended at 96 - the geometric mean was -2.02%. The gap, 0.0202, matched half the variance of 0.02: the volatility drag, geometric equals arithmetic minus half the variance.

3

Adjust before you compare

Real returns strip out inflation - 8% over 3% is 4.85%, not 5%; excess returns subtract the risk-free rate. Choose log returns to model and sum, simple returns to compound a portfolio, and always average with the geometric mean.

Next — Part 2 practice: the stylized facts of returns, from fat tails to volatility clustering.